



SAFETYVISION

PPE Violation Detection

Automated construction-site safety monitoring, powered by YOLO11 and Streamlit

Detects missing helmets and vests in video, blurs violators for privacy, and flags sustained violations automatically.

YOLO11

ByteTrack

Streamlit

Roboflow

The problem

Personal protective equipment saves lives — but watching for compliance doesn't scale.



1

Manual monitoring doesn't scale

A single safety officer can't watch every camera feed on a large site, all shift long.



2

Human attention drifts

Reviewing hours of footage for brief violations is tedious, and fatigue causes misses.



3

Feedback arrives too late

Violations are often noticed only after an incident report, not while they're happening.

Dataset

Roboflow — "helmet-vest-detection" (version 10), 4 annotated classes



Split overview

352

training images

100

validation images

48

test images

Classes



complete_vest_helmet

Fully compliant worker



no_helmet

Helmet missing



no_vest

Vest missing



no_vest_no_helmet

Both missing

Model and training

YOLO11s fine-tuned on the PPE dataset



YOLO11s

9.4M parameters • 101 layers (fused) • 21.3 GFLOPs



50

Epochs



16

Batch size



640 × 640

Image size



AdamW

Optimizer

Initial learning rate: 0.01

Test set results

Evaluated on 48 held-out test images



0.805

Precision



0.810

Recall



0.891

mAP@0.50



0.650

mAP@0.50-0.95

Precision and recall balance false alarms against missed violations; mAP@0.50 reflects strong detection at a standard overlap threshold.

How the app processes a video

Frame sampling keeps the model from analyzing every frame, trading a small, tunable delay for speed.

01



Upload

A video file is uploaded through the Streamlit interface.



02



Sample

Every few seconds, a short burst of frames is sent to the model — the rest are skipped to save compute.



03



Detect & track

YOLO11 finds PPE violations; ByteTrack keeps a consistent ID per person.



04



Blur & flag

Frames with a violation are blurred, with only the violator's box left sharp.

App controls

Streamlit sidebar settings, tunable per video



Confidence threshold

Minimum detection confidence the model must reach before a box counts.



Blur strength

How strongly the frame is blurred around a detected violation.



Test interval

How often (in seconds) the model runs a fresh detection pass.



Frames per burst

How many consecutive frames are analyzed each time the model runs.

Limitations & future work

Known constraints today, and where the project is headed next



Lighting & occlusion

Accuracy drops in poor lighting, heavy shadows, or when a worker is partially blocked from view.



Limited PPE classes

Only helmets and vests are covered today; gloves, boots, and harnesses aren't detected yet.



Not yet real-time

Frame sampling trades a small delay for speed; live camera streaming isn't supported yet.



Multi-camera & alerts

Planned: fusing multiple camera feeds and sending automatic alerts to site supervisors.



Thank you

Team



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